Assessing the Importance of Sex and Disease-Specific Anatomy in Electrophysiology and Mechanical Simulations with a Newly Developed Public Virtual Cohort of Four-Chamber Heart Models

José Alonso Solís-Lemus[1,◕], Rosie K. Barrows[1,◕], Cristobal Rodero[1], Marina Strocchi[1,6], Natalie Montarello[2], Nishant Lahoti[2], Cesare Corrado[1], Abdul Qayyum[1], Shahrokh Rahmani[1], Caroline Roney[3], Gernot Plank[4,5], Cristoph Augustin[4,5], Hao Xu[2], Alistair Young[2], Pras Pathmanathan[7], Ronak Rajani[2], Steven A. Niederer[1,8]

1 National Heart and Lung Institute, Imperial College London, UK
2 School of Biomedical Engineering and Imaging Sciences, King's College London, UK
3 Queen Mary University of London, UK
4 Gottfried Schatz Research Center, Division of Medical Physics and Biophysics, Medical University of Graz, Austria
5 BioTechMed-Graz, Austria
6 Cardiac Rhythm Management, Medtronic, London, UK
7 Food and Drug Administration, United States
8 Alan Turing Institute, London, UK

◕These authors share first authorship.
* j.solis-lemus@imperial.ac.uk

Abstract

This work presents a study on how differences in cardiac anatomy attributed to sex and disease can influence cardiac electrophysiology and mechanics using a virtual cohort of four-chamber heart models. Patient anatomy varies across sex and disease. However, capturing this variation in in-silico studies remains poorly accounted for, with studies often using either single representative cases or imbalanced virtual cohorts. Whole-heart electromechanics models incorporate the patient's anatomy, electrophysiology and mechanics across different scales, from molecular, tissue and whole-heart and circulatory system levels. However, cardiac models are typically built from one or a small number of anatomies, with sex rarely reported and the effects of anatomical variability, which include those due to sex or disease, largely unexplored. This limits clinical translation and reduces regulatory credibility.

We developed fifty patient-specific anatomical models of 25 male and 25 female hearts in heart failure and control cases. We ran benchmark passive inflation and paced activation simulations with consistent parameters and boundary conditions across cases to isolate the impact of anatomical variations with sex and disease.

Heart failure models exhibited increased chamber volumes, larger volume changes during inflation, and delayed activation times relative to controls. These trends were consistent across sexes, although right ventricular activation showed a significant sex-based difference.

Variations in anatomy with sex and disease have a significant impact on cardiac simulations, which support the inclusion of multiple heart anatomical models in in-silico trials. The resulting virtual cohort captures key anatomical variability and is publicly available at `https://doi.org/10.5281/zenodo.19351401` and our code is available at `https://github.com/OpenHeartDevelopers/cemrg-heartbuilder`.

Author summary

Understanding how sex and disease affect the heart's electrical and mechanical behaviour is essential for developing accurate and credible computational models. In this study, we built a balanced virtual cohort of fifty patient-specific, four-chamber heart models from male and female individuals with and without heart failure. We designed a single, streamlined pipeline to generate these models and ran benchmark simulations of electrical activation and passive inflation to evaluate how anatomical differences influence cardiac function.

We found that heart failure models showed larger chamber volumes, greater volume changes, and delayed activation times compared with controls. These patterns were consistent between sexes, although right ventricular activation showed a clear sex-related difference. Our results demonstrate that anatomical variability across sex and disease has a measurable effect on cardiac simulations. By releasing this virtual cohort and its underlying pipeline publicly, we aim to support reproducible and transparent in-silico studies and to advance the use of virtual populations in regulatory and clinical applications.

1 Introduction

Patient-specific computational models and simulations of the heart have increasingly been used to develop and guide clinical therapies [1], supporting the design [2, 3], evaluation [4, 5], and delivery [6, 7] of innovative clinical therapies. These models could advance personalised medicine by enabling the development and in-silico testing of therapies tailored to the anatomical and physiological features of individual patients. However, the creation of credible and trustworthy cardiac models remains challenging [8, 9].

There is growing support for the use of simulations for regulatory submission by the Medicines and Healthcare products Regulatory Agency [10], European Medicines Agency [11], and U.S. Food and Drug Administration (FDA) [9, 12]. In particular, the FDA has identified scientific gaps and challenges that have limited the credibility of simulation models intended for medical use, including (i) insufficient analytic methods, (ii) lack of established credibility assessment tools, (iii) absence of best-practice guidelines, and (iv) limited availability of high-quality, comprehensive data sets [12].

Data scarcity remains a fundamental obstacle to the development of robust cardiac models, particularly the lack of high-quality, sex-balanced, disease-specific patient data. Several studies have attempted to bridge this gap through the creation of virtual cohorts that can be used for in-silico trials [13, 14]. Despite these advances, addressing sex balance in cardiac modeling databases is a persisting challenge. The cohort presented by Strocchi et al. [13] has one female patient, while the one presented by Rodero et al. [14] has 6 out of 20. Finally, in the study by Roney et al. [15] on predicting atrial fibrillation recurrence, 28 of the 99 patients were female. Large virtual heart cohorts have begun to emerge, enabling in-silico trials and population-level analyses [16–18]. For example, the Strocchi et al. [13] database has been widely used for studies on valve mechanics [19], fibrosis detection [20], and arrhythmia modeling [21]. While these cohorts represent valuable tools for the cardiac modelling community, they also highlight the challenge of having balanced representation of sex or disease subtypes, and workflows for generating new whole-heart cohorts remain poorly defined.

Sex bias is a complex, multi-factorial problem that affects the availability and generalisability of data used for computational models and simulations [22]. Simulation studies can take considerable time and computational resources, which makes it especially relevant to consider whether anatomical differences associated with sex or disease influence the results. The impact of such variation on simulation outcomes as well as on cardiac function is not frequently explored or understood. Understanding this relationship is key to determining when sex-specific anatomical data should be included in model development and evaluation.

Testing the importance of anatomical variability requires creating sex balanced virtual cohorts. However, building a model of the whole heart is a labour-intensive and technically demanding task. Available solutions involve multiple stages: image segmentation, mesh generation, and the assignment of fibre orientations. Some approaches have explored the use of deep learning to automate aspects of mesh generation [23]. However, these methods are typically limited to single cardiac chambers rather than full four-chamber models.

CemrgApp [24] is an open-source medical imaging platform with image processing toolkits for cardiovascular research, that has a low barrier to entry and low learning curve, putting reproducibility and best practices at the forefront. CemrgApp has enabled semi-automated creation of high-quality left atrial models and has been employed in studies involving atrial fibrosis characterisation [25], motion quantification [26], and regional strain analysis [27]. However, the extension of such tools to comprehensive whole-heart modelling, including all four chambers, has not been systematically demonstrated or standardised.

In this study, we developed CEMRG Heartbuilder, a Python-based library designed to systematically generate patient-specific, four-chamber heart models from clinical computerised tomography (CT) data. We applied the workflow to a new cohort of 50 patients (balanced by sex), classified into three clinical groups: controls ($n = 26$), heart failure with narrow QRS ($n = 12$), and heart failure with wide QRS ($n = 12$). For each case, models were created from a CT scan to a simulation-ready mesh with fibres, and subsequently used to run benchmark electrophysiological and mechanical simulations. The simulations demonstrated the feasibility and robustness of the approach, and, through the use of uniform material properties, isolated the impact of disease and sex differences in anatomy on reference mechanical and electro physiological simulations. Figure 1 provides an overview of the pipeline and the resulting cohort.

2 Study population

Fifty cases were processed, creating patient-specific four-chamber heart models. The cohort was split into three groups: Controls (C, $n = 26$), Heart Failure (HF) with narrow QRS (HF_N, $n = 12$), and HF with wide QRS (HF_W, $n = 12$); each group is balanced by sex (50% Female). A QRS duration of less than 120 ms is considered narrow. Each case consisted of a CT scan, acquired using a consistent imaging protocol. The reconstructed meshes represent the cardiac anatomy in a static reference configuration. Subsequent mechanical simulations apply prescribed endocardial pressures from this baseline geometric state, enabling comparison of anatomical effects across the cohort. The study was conducted under ethical approval titled Retrospective analysis of cardiac imaging datasets for development of novel biomarkers study (Short title: RAIDER), approved by GSTT, reference number 306914, protocol version 1.0 01/10/2021. A summary of the cohort's demographics is presented in Table 1. Summary statistics include the number of participants per group, age, ejection fraction (EF) as a percentage, and QRS duration in milliseconds (ms). Male control cases were patients with no known cardiac disease or symptoms. Five cases had no echocardiography data available and 1 case had no ECG.

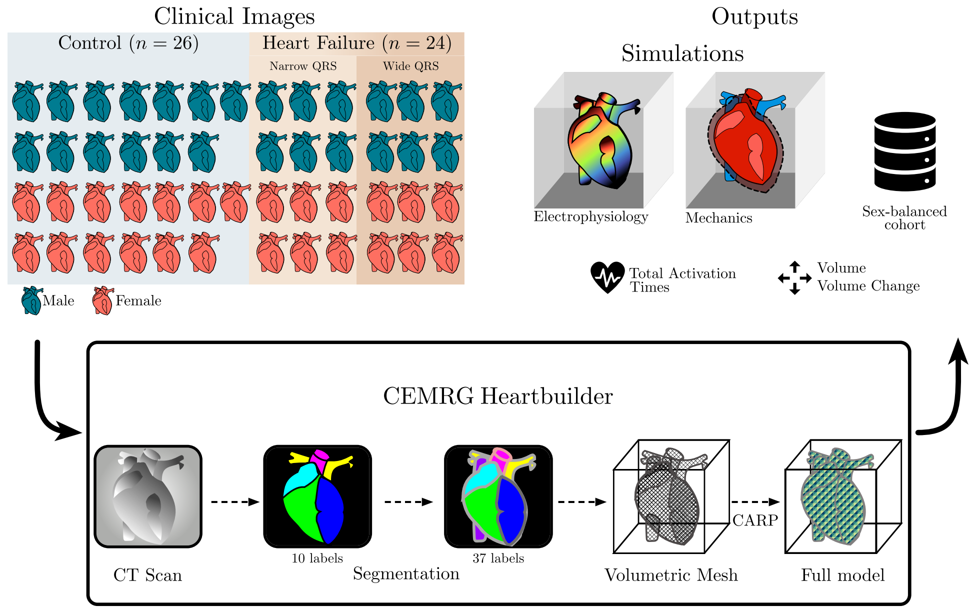

Fig 1. Cohort creation and pipeline overview. Starting from clinical CT scans, the workflow performs multi-label segmentation and mesh generation to create simulation-ready models with assigned fibre orientations. A new dataset of 50 patients (26 controls, 24 with heart failure subtypes: narrow QRS and wide QRS) balanced by sex is processed. Resulting models are used in simulations of cardiac electrophysiology and mechanics to extract clinically relevant measurements.

Table 1. Summary of the 50-patient cohort demographics, grouped by sex and condition. See text for full description of the cohort and the conditions. Abbreviations: M=Male, F=Female, HF=Heart Failure, HF_N=HF with Narrow QRS duration, HF_W=HF with Wide QRS duration.

	Total	Age	EF (%)	QRS duration (ms)
M_C	13	53.4 ± 12	$58.7 \pm 4(n = 8)$	$90.3 \pm 8, (n = 12)$
M_{HF_N}	6	53.8 ± 13.8	34.4 ± 14.2	103 ± 7.8
M_{HF_W}	6	67.3 ± 16	38.5 ± 8	163.8 ± 18
F_C	13	49.5 ± 9.8	58.8 ± 4.3	87 ± 12.76
F_{HF_N}	6	61 ± 11.4	43.6 ± 7.1	88.3 ± 9.6
F_{HF_W}	6	59.2 ± 7	36.1 ± 7.8	144.7 ± 13.3

3 Methods

We developed CEMRG Heartbuilder, a pipeline for the streamlined creation of simulation-ready meshes, generated from a CT scan. The pipeline to process the CT scans is based on the work by [13] but streamlined into a single integrated workflow. Processing of the information consisted of three stages: image to mesh analysis, consisting of a multi-stage segmentation (Section 3.1.1), upsampling, and mesh generation (Section 3.1.2); mesh processing and model creation, which includes the assignment of fibre orientations in the ventricles and atria, and the tagging of the mesh with the different labels needed for the simulations (Section 3.2); and simulations. The multi-stage segmentation and mesh extraction substages are open source, whilst some of the mesh post-processing and simulation stages use the cardiac arrhythmia research package (CARP) [28]. Note that the modular structure of the CEMRG Heartbuilder allows for third-party projects to be included to be made completely open source.

3.1 Image to Mesh Stage

3.1.1 Multi-stage Segmentation

The image to mesh stage consists of a segmentation of the CT scan via the deep learning model by [29], Figure 2a, which identifies 10 distinct regions: left ventricle myocardium, and blood pools of the left and right ventricles, the left and right atrium, the pulmonary artery, the aorta, the left and right atrial pulmonary veins, and left atrial appendage. Regions are marked in the segmentation with different integer values, we refer to these as "labels" throughout the manuscript. The user then identifies the different labels in the U-Net output, which splits the pulmonary veins into the superior and inferior veins. The user selects two sets of 3 points to identify the location and orientation of the superior and the inferior venae cavae. A post-processing stage follows, increasing the number of labels to 37 (Figure 2b). This stage creates the myocardia, valve planes, and vein rings. The myocardia of the right ventricle (RV), the atria, the aorta and pulmonary artery are extracted through the use of distance maps, with a user-selected thickness of 3.5mm for the RV, and 2mm for the remaining structures and coupling the different regions to ensure no overlap occurs. A table of the labels is provided in Supplementary Table S3.

CemrgApp aggregates the different workflows through a combination of code running natively on the app and a docker wrapper, which calls the segmentation and different post-processing stages. The image analysis stage finalises by upsampling the 37-label segmentation to an isotropic resolution of 0.15mm, and a applies a smoothing algorithm [30].

3.1.2 Conversion to Mesh

A volumetric mesh is extracted from the smooth segmentation using a meshing tool developed in-house, which comes packaged with CemrgApp and is based on the Computational Geometry Algorithm Library (CGAL) [31]. The CGAL mesher generates unstructured volumetric meshes composed of linear tetrahedral elements (4 nodes per element). Target element edge length was set to approximately 0.5 mm via the `cell_size` and `facet_size` parameters. Mesh density is controlled via CGAL parameters (`facet_size`, `cell_size`, `facet_distance`), which are specified globally for the entire heart geometry. Complete parameter specifications are provided in Supplementary Table S2. The next step is to extract the myocardia, simplify the mesh topology, and relabel the tags using meshtool [32]. The simplification consists in identifying elements insufficiently connected to neighbours of the same class. A complete description of the CGAL parameters, which can be modified by the user, is

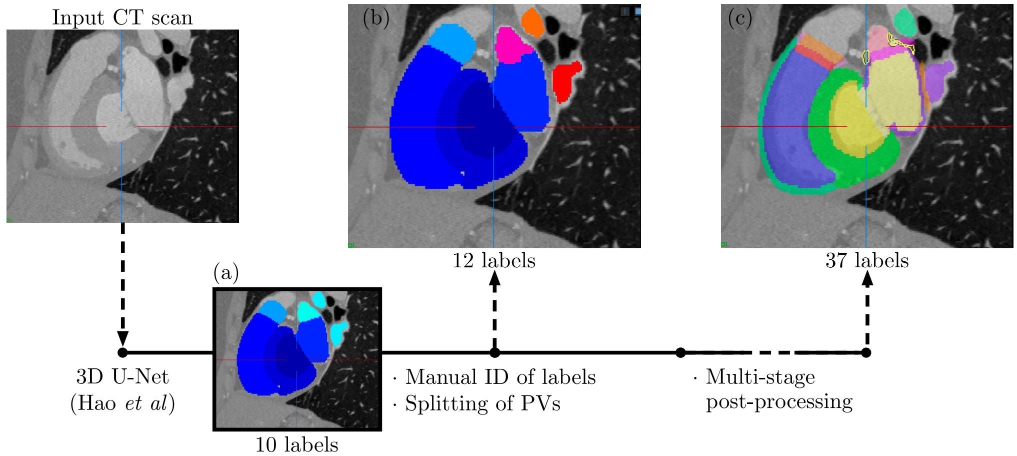

Fig 2. Multi-stage segmentation process. CemrgApp uses the deep learning model developed by [29] to segment the heart. (a) The output of the U-Net model, which identifies 10 regions, or "labels", as described in the text. (b) The output of the intermediate stage, where the user manually splits the pulmonary veins into superior and inferior (marked by contrasting colours). (c) The final output of the post-processing stage, where the myocardium of the different structures is extracted, increasing the number of labels to 37. The final segmented images are then upsampled to an isotropic resolution of 0.1mm and smoothed.

available in Supplementary Table S2. Figure 3 shows the output of the meshing process from the smooth segmentation to the working mesh re-labelled and with the blood pools removed. The final simulation-ready models contained a mean of 2.6×10^6 elements per heart (range 1.9–4.5 $\times 10^6$).

3.2 Mesh Processing and Model Creation

The last stage of the pipeline included takes the smooth mesh and produces a model with fibre orientations, tagged with the different labels needed for the simulations. This stage is run using CARP and meshtool [28, 32] along with the CEMRG Heartbuilder python library. Figure 4 shows a visual overview of the model creation stage, which is described in detail below.

Universal Ventricular Coordinates (UVCs) are a system of coordinates that describe the geometry of the ventricles [33]. Additionally to the ventricles, UVCs are also calculated for the atria, to provide a system of coordinates defined on the volumetric mesh that facilitated the assignment of different tags within the mesh, and of boundary conditions for the pericardium. To achieve this, the left and the right atria were treated as an upside down single ventricle, following the approach described by Strocchi et al. [34], with the apex manually placed between the two right pulmonary veins and behind the supervior vena cava, respectively.

Atrial fibres are then assigned to the mesh by projecting a rule-based fibre field [35] onto the mesh using Universal Atrial Coordinates (UACs) [36]. Ventricular fibres are then assigned to the mesh using a rule-based algorithm, described by [37], which uses a series of Laplace solutions to assign the fibre direction. UACs are a system of coordinates that describe the geometry of the atria [36]. The atrial fibres, which originally are produced on surfaces, are then projected onto the 3D mesh, by assigning endocardial and epicardial fibres in the inner and outer 50% across the thickness of the atria, respectively. From the UAC pipeline, which is run through a docker container, similar to the work by [1], the sino atrial node (SAN) is also mapped from the atlas.

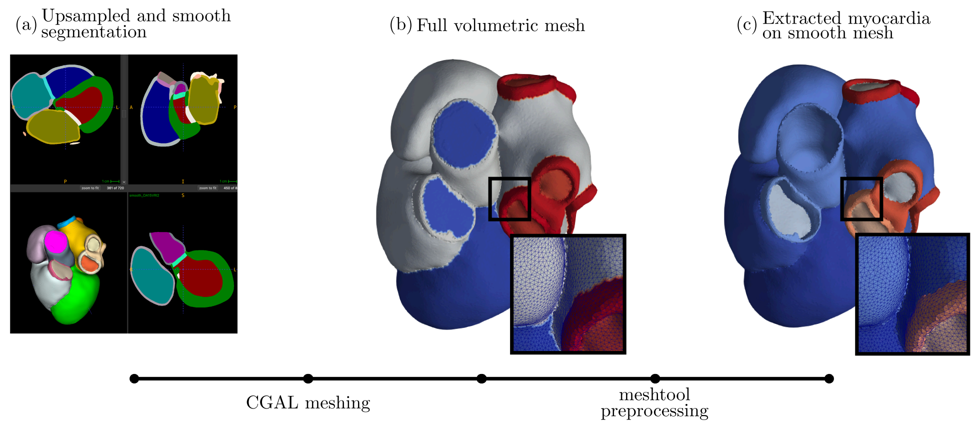

Fig 3. Output of the meshing process from the smooth segmentation to the final working mesh. (a) The upsampled and smooth segmentation. (b) The mesh extracted from the segmentation using the CGAL meshing tool. It contains all the blood pool and myocardium of the different structures. (c) The final mesh after the relabelling, cleaning, smoothing, and extracting only the myocardia and valve planes. Other outputs are created from this final process, such as specific endocardial and epicardial surface meshes for the atria, which will be used later in the process.

The mesh is tagged with the different labels needed for the simulations. First, a region representing the fast-conducting Bachmann bundle is created, to simulate fast propagation of the electrical signal from the right to the left atrium. This is done using the UVCs defined on the atria. The fast endocardial conduction layer (FEC) is also created at this stage, which is a thin layer of fast conducting tissue at the endocardium of the ventricles representing the Purkinje network. In addition, to prevent unphysiological propagation from the atria to the ventricles and to control the atrioventricular delay, the atria and the ventricles were electrically isolated from each other by defining a layer on non-conducting tissue between the atrial and ventriclar myocardium. Finally, the SAN and the left and right ventricular fascicles are also created, Figure 4 (f.1, f.2, f.3). The SAN and fascicles are used to initiate electrical activation in the electrophysiology (EP) simulations. The left and right fascicles represent the first breakthrough from the Purkinje network into the myocardium to mimic the activation reported by the Durrer maps [38]. These locations are defined in UVCs based on [39].

Mesh Quality Assessment We assessed the quality of all 50 volumetric meshes using the volume-based distortion metric (`tet_qmetric_volume`) implemented in meshtool [32], which quantifies element distortion relative to an ideal tetrahedron. This metric is directly related to the determinant of the Jacobian of the geometric map and is normalised to $[0, 1]$, where 0 corresponds to a perfect tetrahedron and 1 to a fully degenerate element. Across all 50 meshes (mean 2.6×10^6 elements per mesh, range 1.9–4.5×10^6), the mean element quality was 0.153 ± 0.100 (mean $\pm$ SD across all elements in the cohort). The minimum quality per mesh averaged 2.1×10^{-4}, indicating that even the worst elements retained near-ideal geometry. No inverted elements (quality > 0.99) were detected. Near-degenerate elements (quality > 0.90) were found in 28 of 50 meshes, comprising at most 9 elements per mesh ($< 0.0003\%$ of total elements). These results confirm high geometric fidelity suitable for finite element analysis.

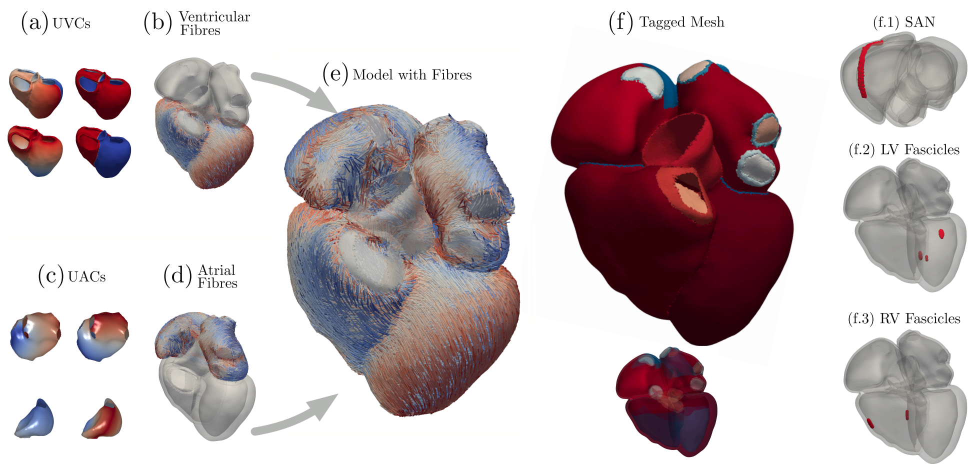

Fig 4. Model creation stage. Universal Ventricular coordinates (a) are calculated from the mesh, then used to assign the fibre direction (b) using a rule-based algorithm [37]. Universal Atrial coordinates (c) are calculated from the mesh, then used to assign the fibre direction (d) using a projection from an atlas [36]. The atrial fibres, which originally are produced on surfaces, are then projected onto the 3D mesh. The fibres from the ventricles and atria are then assigned to the model (e). The final mesh is tagged with the different labels needed for the simulations (f). The fast endocardial conduction layer (FEC) is also created at this stage (f - bottom), which is a thin layer of fast conducting tissue at the endocardium of the ventricles. Finally, UVCs and UACs are used to create the sino atrial node (SAN, f.1), the left ventricle fascicles (f.2) and right ventricle fascicles (f.3).

3.3 Simulations

We performed EP and mechanics simulations on all the models. Simulations were run with a fixed set of parameters to assess the impact anatomical differences would have on the outputs. These simulations also served as an additional quality check for the meshes, to ensure the models could be used to run electromechanics simulations.

Each patient-specific mesh was simulated in its native coordinate system without spatial registration to a common anatomical reference frame. Comparisons between models were performed on scalar outputs (chamber volumes, total activation times) that are invariant to rigid transformations. Universal Ventricular Coordinates (UVCs) [**?**] provide normalized transmural and apicobasal coordinates for regional functional analysis independent of absolute spatial positioning.

The EP simulations were started at the SAN, defined in Section 4. The ventricular endocardial electrodes (fascicles) were also activated simulating the early activation sites [40]. The reaction-eikonal model [41] was solved to compute the activation times at each node. A conduction velocity of 1 m/s was used at the myocardial tissue, with a 40% anisotropy cross-fibre, and a 6-fold speed in the fast activation regions [42]. As output, the total activation time of each ventricle, of both ventricles and both atria were extracted.

To test how differences in anatomy impacted simulation predictions, we performed inflation simulations to evaluate the mechanical behaviour of the models. Briefly, increasing pressure is applied at the endocardium of each chamber until a maximum established pressure of 7 mmHg for the left chambers (LV and LA) and 3.5 mmHg for the right chambers (RV and RA) is reached. The myocardium was modelled as an transversely isotropic hyperelastic material using the Guccione's material law [43], while the rest of the cardiac structures were modelled as isotropic hyperelastic materials using a NeoHookean material law [44]. Mechanics simulations were run with the same boundary conditions as the work by [45], which can be consulted in the Supplementary Material.

Simulations were run to assess the impact of anatomical differences on the outputs and to serve as a quality chech for the meshes. All simulations were run with CARP [28]. The EP simulations were run in a 20-core workstation, while the inflation simulations were run in the UK national supercomputer ARCHER2, using 512 cores.

Model outputs obtained from the simulations are listed in Table 2, along with a brief descripton. The 18 measurements are grouped into four categories: activation times (4 outputs), inflated volume (4), volume change (absolute 4, and normalised 4), and volume ratios (2). The absolute volume change (ΔVol) is calculated as the difference between the maximum and minimum volume during the inflation simulation. For simplicity, we refer to inflated volume as volume throughout the manuscript. The normalised volume change ($\mathcal{V}$) is calculated as the percentage of change in volume, and the volume ratio is calculated as the ratio between the left and right volumes, this was calculated for the ventricles and the atria. The Total Activation Time (TAT) is the time it takes for the electrical signal to propagate through the entire chamber.

QRS duration. Since the simulations do not include a torso, the comparison cannot be made directly with the QRS duration measured in the ECGs. Instead, we used the total activation time in the ventricles as a surrogate for QRS duration and compared it between the different groups.

Total Activation Time in the Atria. This parameter has emerged as a key marker of atrial remodeling and is associated with HF progression and complications [46, 47]. We assessed the total activation time in the atria, comparing it between the different groups.

Table 2. Model outputs obtained from the simulations. The subscript x refers to the chamber, which can be left ventricle (LV), right ventricle (RV), left atrium (LA), or right atrium (RA).

Measurement	Symbol	Calculation / Definition
Total Activation Time	TAT_x	All chambers, ventricles, and atria
(Inflated) Volume	Vol_x	All chambers
Volume Change	$\Delta\mathrm{Vol}_x$	Maximum minus minimum volumes
Normalised Volume Change	$\mathcal{V}_x$	Percentage of change in volume
Volume Ratio	$\frac{L}{R}$	Ratio of left and right volumes

3.4 Statistical tests and comparison of volume to clinical literature

We performed a multi-step statistical analysis of our cardiac dataset. First, we computed descriptive statistics (mean ± SD) for all core metrics (volumes and activation times) stratified by sex, condition (HFvs. control), and subtype (HF_Nvs. HF_W). Differences between narrow and wide QRS durations are only explored qualitatively, as the number of cases in each group is low.

Effect sizes and statistical significance. To assess the differences in the outcomes, we used a two-way ANOVA to test the effects of Sex, Condition, and their interaction. We followed up significant ANOVA results in the interactions with post-hoc pairwise comparisons using Tukey's method, and calculated Cohen's d effect sizes [48] for each comparison. If the interaction was not significant, we performed marginal comparisons between HF and controls, or male and female groups, across all cases. P-values were corrected for multiple comparisons using the Benjamini–Hochberg false discovery rate [49]. Effect sizes are reported as Cohen's d, using Gignac's expanded criteria [50], where $d < 0.5$ is small, $0.5 < d < 0.8$ is moderate, $0.8 < d < 1.2$ is large, and $d > 1.2$ is very large.

To externally validate the models, we compared different outputs from the pipeline linked them to a clinically relevant metrics.

Left atrial and left ventricular volume. Left atrial volume is a predictor of outcome in patients with chronic HF, independently of their symptoms, age, or LV function [51]. LV volume is a key prognostic factor in HF, as it is associated with the severity of the disease and the risk of adverse events [52, 53]. We assessed the left atrial volume (Vol_{LA}) and LV volume (Vol_{LV}) in the models, comparing between the different groups. For Vol_{LV}, we compared measurements from our cohort with reference values from the UK biobank (UKBB) [54], which defines normal ranges for left and right ventricles in males and females. Outside the normal ranges, we considered the volumes to be abnormal, and therefore indicative of HF or other cardiac conditions.

4 Results

Fifty whole-heart models (Figure 5) were created, each including EP and mechanics simulations described in Section 3.3. All simulations were performed with the same parameters for each model. Therefore, the differences in simulations between sexes reported below relate to differences in anatomy. We first examined the distributions of all core metrics across sex and condition. A full table of means and standard deviations is provided in the Supplementary Material. We found no significant difference in age between males and females (M: 53.4 ± 15.0 y; F: 54.6 ± 10.7 y; $p = 0.75$).

Geometric characterization and correlation with simulation outputs. Geometric metrics stratified by sex and condition are provided in Supplementary Table S1. Heart failure patients exhibited substantially larger chamber volumes compared to controls, with male HF patients showing the most pronounced dilation (LV: 289 ± 88 mL vs control: 180 ± 36 mL, $p < 0.001$). Myocardial wall volume scaled with anatomical size, ranging from 263 ± 37 cm^3 in female controls to 396 ± 75 cm^3 in male HF patients. Geometric variables showed strong correlations with simulation outputs (Supplementary Figure S2). Chamber volume was the strongest predictor of volume change during passive inflation ($\rho = 0.96$ for LV, $p < 0.001$), indicating that larger ventricles exhibit greater absolute compliance under uniform material properties. Myocardial wall volume correlated strongly with activation times ($\rho = 0.87$ for $\mathrm{TAT}_{\mathrm{RV}}$, $\rho = 0.85$ for $\mathrm{TAT}_{\mathrm{V}}$, both $p < 0.001$), reflecting the expected increase in conduction path length with anatomical size. Notably, cross-chamber geometric correlations were observed, such as RV volume predicting LV activation time ($\rho = 0.81$, $p < 0.001$), consistent with whole-heart anatomical coupling.

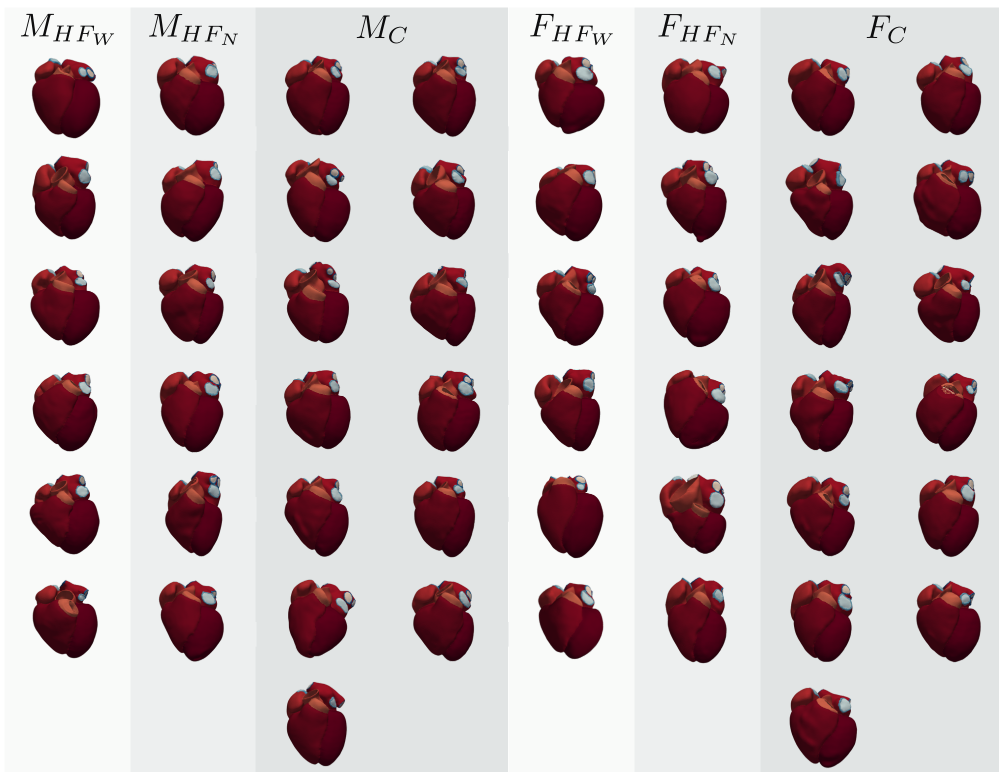

Fig 5. Three-dimensional whole-heart reconstructions from the virtual cohort, grouped and labelled by sex and condition. Each colour corresponds to a different structure in the mesh. Each model was derived from a patient's CT scan and includes fibre orientations, illustrating anatomical variability across sex and heart-failure subtypes. Abbreviations: M_{HF_N}: Male, HF narrow-QRS; M_{HF_W}: Male, HF wide-QRS; M_C: Male, Control; F_{HF_N}: Female, HF narrow-QRS; F_{HF_W}: Female, HF wide-QRS; F_C: Female, Control.

Next, we assessed model volumes and the outputs of the EP and mechanics simulations. In each case, we report the results from the two-way ANOVA and the post-hoc tests, showing the effect sizes. Where no interaction was present, we examined independent effects. In both cases, only statistically significant results after FDR correction are reported here. Effect sizes were computed using Cohen's d and are visualised in Figure 7.

LV volume. Vol_{LV} was significantly larger in HF patients and showed a significant interaction ($p < 0.05$ in ANOVA and all post-hoc comparisons). For all but three male controls, the LV volumes fall within the normal range ($[109, 218]$ mL) compared to the UKBB reference values [54]. Similarly, female controls also fall mostly within their corresponding normal range ($[88, 161]$ mL). We display the results in Figure 6. The effects sizes were also large between HF and control groups, but they were much larger in males than in females, for instance, interactions involving females ranged between $0.9 < d < 1.17$, while interactions involving males were higher than $d > 1.4$ ($p < 0.05$ in all cases, Figure 7(a)).

Fig 6. Comparison of measured ventricle volumes from the virtual cohort with clinical literature values and ranges from UKBB. The UKBB normal ranges are defined as measurements within the 95% prediction interval of the study by [54]. Two graphs, corresponding to the LV and RV metrics and reference volumes. Each plot is separated by sex and coloured by condition, with Controls in blue and Heart Failure in orange. Normal ranges are shown as greyed areas, and exceeding the upper limit of the reference range may indicate pathological ventricular dilation. Percentages of each group within the UKBB normal range are displayed in the legend.

Volume change in the LV. We used volume change as a surrogate for stiffness. LVvolume change was significantly larger in males compared to females and in HF patients compared to controls (both main effects with $p < 0.01$), but we did not detect a significant sex $\times$ condition interaction ($p = 0.44$), indicating that sex and condition contribute independently to LV volume change. The effect sizes were large, with $d = 0.94, p < 0.01$ for HF vs controls, and $d = 0.78, p < 0.01$ for male vs female comparisons (Figure 7(b)).

Activation times. For ventricular TAT_V, no significant interaction was found between sex and condition ($p = 0.76$), however, the independent effect sizes were large. First, TAT_V was significantly longer in males compared to females ($d = 1.4, p < 0.01$), and in HF patients compared to controls ($d = 0.72, p = 0.014$). TAT_{LA} showed a similar pattern. First, no significant interaction between sex and condition was found ($p = 0.24$). Second, the independent effects were significant and moderate, showing prolonged activation times in HF vs. controls ($d = 0.62, p < 0.05$) and prolonged activation in males vs. females ($d = 0.59, p < 0.05$).

To assess the validity of using TAT as a surrogate for QRS duration in the absence of torso modeling, we correlated simulated ventricular TAT with recorded ECG QRS duration in cases where ECG data were available ($n = 33$ of 50). In control hearts, TAT showed a positive correlation with QRS duration (Pearson $r = 0.649, p < 0.005$, Supplementary Figure S1), indicating that simulations using reference conductivity parameters provide an approximate anatomical surrogate for QRS duration, and that anatomical variation captures at least part of the inter-subject variability in activation time. In heart failure patients, no correlation was observed ($r = -0.127, p = 0.64$), consistent with the expected influence of pathological tissue properties (fibrosis, conduction blocks) that are not represented in our standardised parameter set.

Normalised volume changes in LV. $\mathcal{V}_{LV}$ showed a significant interaction in the ANOVA between sex and condition ($p < 0.05$). Post-hoc tests only showed effects sizes were significant when comparing

Male HF vs. male controls ($d = -1.33$),

Male vs. female with HF ($d = -1.09$), both with $p < 0.05$.

5 Discussion

We created a sex-balanced virtual cohort of 50 whole-heart models derived from clinical CT scans of both healthy individuals and HF patients. We used a semi-automated pipeline to generate these models, which includes segmentation, mesh generation, fibre orientation assignment, and simulation setup. The models enabled the exploration of differences in volume, and activation time across sex and disease groups, using benchmark electrophysiology and mechanics simulations. With this study, we demonstrate that anatomical differences between sexes and across disease states have a significant impact on both electrical and mechanical simulation predictions in the heart. This highlights the critical need to account for such variations when developing and applying computational models in cardiology.

A variety of computational pipelines have been proposed to construct heart models, ranging from atria-only or ventricles-only representations to anatomically detailed four-chamber reconstructions. The works by [55] and [56] proposed patient-specific atrial models, including anatomical regions and fibre orientations from clinical data. In contrast, other pipelines focus only on ventricular chambers [57, 58]. A notable whole-heart model was developed by [59], which builds detailed four-chamber meshes of the entire heart from patient imaging and has been enhanced since to improve simulation fidelity, multi-scale integration, and clinical translation. The resulting models are anatomically detailed, but the method depended on extensive user guidance to create the initial atlas, limiting throughput and scale. Other groups, such as Gillette et al. [60], have also demonstrated personalised whole-heart electrophysiology models, but again with significant manual setup and typically small cohorts. In summary, existing pipelines either produce atria-only or ventricle-only models, or else deliver high-detail four-chamber meshes at the cost of manual effort. Our framework seeks to overcome these limitations by producing anatomically complete four-chamber (atria + ventricles) models with minimal manual input, combining scalability with high anatomical fidelity.

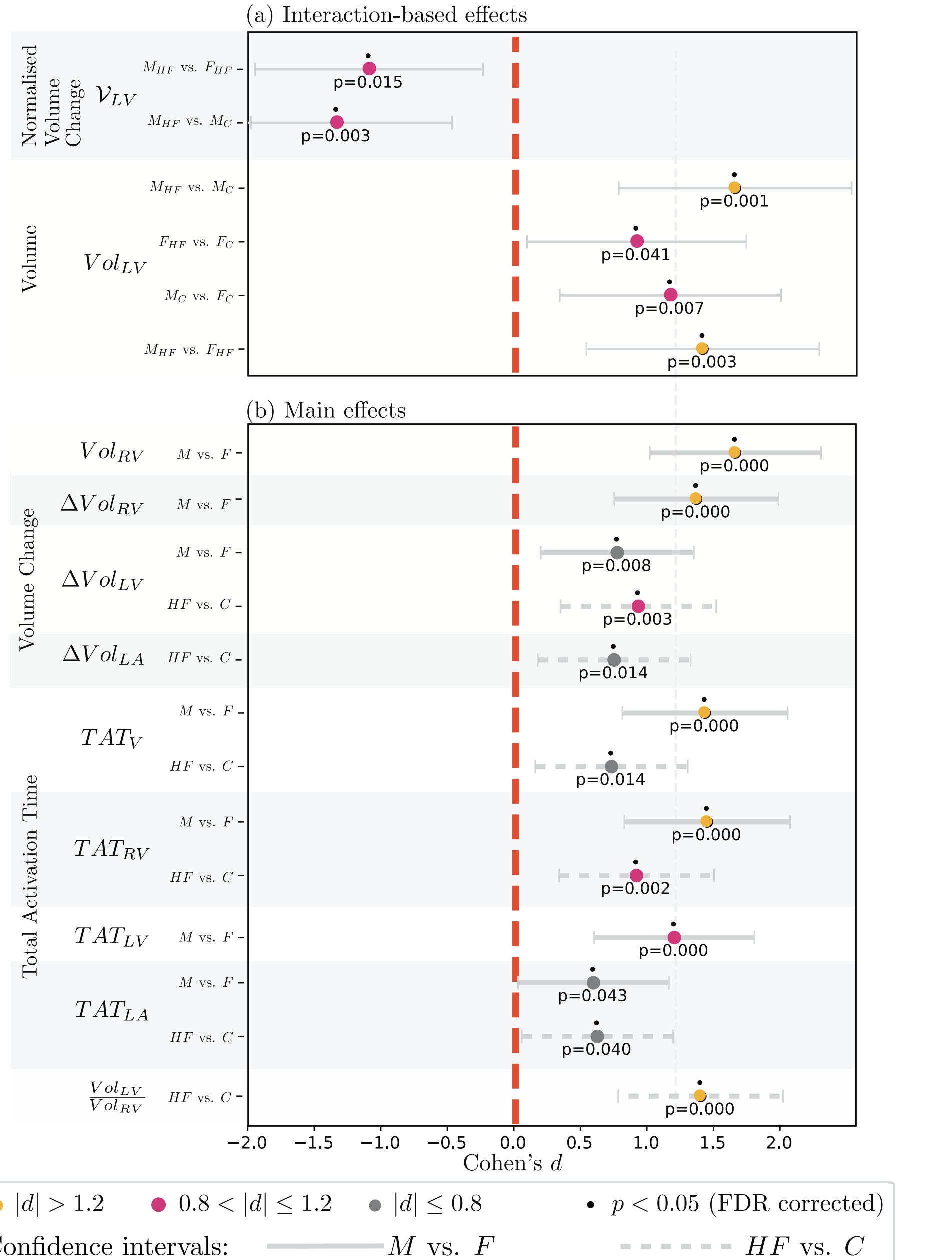

Fig 7. Cohen's d effect sizes from post-hoc comparisons following two-way ANOVA. Metrics and group comparisons are displayed on the y-axis. This graph only shows effects with a high effect size and a corrected p-value $p < 0.05$. (a) Effect sizes for group comparisons performed when the Sex × Condition interaction was statistically significant, justifying 4 simple-effect post-hoc tests. (b) Effect sizes for marginal comparisons, when no significant interaction was found. Error bars line styles represent the comparison made: solid for sex, dotted for condition. *Abbreviations*: M=Male, F=Female; C=Control, HF=Heart Failure; TAT=Total Activation time, L/R=Left/Right, V=Ventricle.

5.1 Building Patient-Specific Whole-Heart Models: Workflow Insights

The models were created using a standardised pipeline, CEMRG heartbuilder, which allows for the generation of patient-specific models from CT scans. The segmentation framework has enabled a substantial reduction in overall model generation time relative to earlier methods. The pipeline requires approximately 5 hours per heart on average. It is also significantly more detailed than most publicly available pipelines. For instance, methods evaluated in the MICCAI 2024 Whole Heart Segmentation Challenge [61] typically produce 7–10 output labels, insufficient for physiologically realistic biophysical simulations. In contrast, our approach employs a 37-label standard [13, 34], encompassing detailed delineation of blood pools, myocardia, valve planes, and major vessels across all four chambers. The current implementation applies uniform meshing parameters globally across the whole heart. While regional refinement could theoretically be achieved by meshing structures separately and stitching at interfaces, we prioritized mesh integrity to ensure numerical stability in coupled electromechanical simulations, where interface discontinuities can introduce artifacts. The strong correlations between geometric metrics and simulation outputs (Supplementary Table S1, Figure S2) demonstrate that anatomical variability is a primary driver of functional differences when using standardized electrophysiological and mechanical parameters. Chamber volume alone explained 92% of variance in volume change during inflation ($\rho = 0.96$), indicating that anatomical size dominates mechanical behavior under uniform constitutive laws. Similarly, myocardial wall volume predicted 76% of variance in right ventricular activation time ($\rho = 0.87$, $p < 0.001$), consistent with total myocardial mass determining conduction path length and duration.

These findings validate our approach of isolating anatomical effects through parameter standardization. However, they also highlight the limitation: real-world patient-specific predictions would require calibration of tissue properties (contractility, stiffness, conduction velocity) to observed clinical metrics.

5.2 Impact of Anatomical Variability on Electromechanical Predictions

All whole-heart models were simulated using identical parameters for both electrophysiology and mechanics. We did not include active contraction in the benchmark as these simulations often require patient specific parameters to match contraction, pre-load and after load, reducing our ability to compare consistent simulations across anatomies. As such, the results reflect the isolated impact of anatomical structure variation due to sex or disease status on predicted electrical activation and mechanical inflation. The results from the two-way ANOVA showed, in most cases, non significant interactions. This means that we could only look at the independent effects: control vs HF and male vs female. The characterisation of effect sizes (small, moderate, and large) by [50], were included as a guideline, based on the distribution of effect sizes obtained.

5.3 Comparison with Clinical Reference Ranges

Comparison with population-based MRI reference data [54] demonstrated that control models reproduced expected sex-specific chamber sizes, serving as an anatomical fidelity check for our reconstruction pipeline. Male and female LV and RV volumes were largely consistent with normal ranges, supporting the accuracy of our segmentation and meshing workflow. However, because mechanical simulations used uniform material properties across all cases, the observed volumes represent anatomical structure rather than patient-specific functional state. In heart failure, LV enlargement was most pronounced in males, frequently exceeding reference thresholds, consistent with classical patterns of systolic dilation. Females also showed increased LV volumes, though these remained mostly within upper limits, highlighting sex differences in disease expression. RV enlargement was less prominent in both males and females with HF, reflecting clinical reports that RV size is less sensitive to early disease [62]. These results demonstrate that the virtual cohort captures both healthy anatomy and pathological remodeling patterns at the population level.

Impact of Sex and Heart Failure in *In-Silico* Studies Our simulations consistently separated groups by sex and disease status using uniform electrophysiological and mechanical parameters, demonstrating that anatomical variability alone produces measurable functional differences. When population-level material properties are applied to pathological geometries, the resulting simulations capture group-level trends consistent with clinical observations. Absolute LV volume increases were particularly marked in male HF, reproducing the clinical signature of pathological dilation [52,53]. Larger ΔVol_{LV} in HF models reflects the mechanical consequence of dilated geometry under uniform compliance parameters [63,64]. Similarly, larger ΔVol_{LA} patterns align with clinical markers of atrial dysfunction [65,66]. When expressed as normalized change (Figure 7(a)), however, HF females showed the greatest relative differences, driven by smaller baseline chamber sizes. This emphasizes that absolute and relative measures provide complementary insights in sex-stratified analyses [22]. Electrical activation followed similar patterns: ventricular and atrial activation were prolonged in HF and in males, consistent with clinical QRS and conduction observations [46,47]. These convergent trends highlight that population-level simulations with anatomically accurate models can reproduce group-level clinical patterns, while individual patient predictions would require tissue-specific parameter calibration [13].

5.4 Establishing Model Credibility and Validation Strategies for In Silico Clinical Trials

In silico studies using virtual cohorts of patients, referred to as in silico clinical trials (ISCTs), require careful demonstration of model credibility for their conclusions to be considered reliable. Credibility is primarily established by performing validation, that is, comparison of model predictions with real-world data. However, unlike relatively simple models of standalone medical devices, or drug safety/efficacy studies using a generic action potential model, ISCTs present a much wider range of possible validation strategies. As discussed in detail in [67], possible evaluation activities include: validation of a device/drug model in isolation (e.g., bench test validation of a pacemaker lead mechanics model); individual-level validation of patient models in the cohort (i.e., comparing predictions from a specific patient model with clinical data from the specific patient the model represents), population-level validation of the cohort or sub-cohorts (e.g., comparing the distribution of model outputs across the cohort with clinical data from the corresponding real-world population), validation of coupled device-patient or drug-patient models (e.g., PKPD validation studies), and validation of any statistical model used to convert patient model outputs into clinical endpoints. Also, there are important additional activities that should be performed, such as assessing the virtual cohort representativeness; these are not strictly model validation activities and more akin to model input validation for simple models.

Overall, an ISCT is expected to be supported by a body of evidence on model credibility. For a virtual cohort such as the sex-balanced 50 patient cohort provided here, intended for re-use in variety of future device or drug ISCTs, a hierarchical evaluation approach can be used to generate an initial body of credibility evidence to support a future ISCT. In this work, we have conducted a preliminary assessment of anatomical representativeness by comparing ventricular and atrial volumes in the male and female cohort subsets with corresponding UK BioBank data. This comparison demonstrates that our reconstruction pipeline accurately captures population-level anatomical distributions, establishing a foundation for future credibility assessments. However, full validation for in silico clinical trials would require additional hierarchical evaluation steps, including patient-specific functional validation, uncertainty quantification, and context-specific device or drug interaction studies [67].

A natural next step would be to perform sex-specific population-level validation: demonstrating that the distribution of model outputs, for each of the male and female sub-cohorts, match the distribution observed in the male and female populations, particularly for outputs for which a statistically significant sex difference was observed. For instance, predicted chamber TAT distributions could be compared with sex-specific population QRS duration data, after an appropriate transformation to account for the fact that TAT is only a surrogate for QRS. Individualized validation, while more challenging, could be performed for newly generated patient models. Ultimately, these results would be supplemented by context-specific validation when the cohort is used in a device/drug ISCT, together with other credibility assessment activities such as verification and uncertainty quantification [68].

5.5 Limitations

This work has four main limitations that should be considered when interpreting the results. First, the small sample size within each category of HF, narrow and wide QRS duration, limits the statistical power for detecting subtle effects or interactions. Nonetheless, by balancing sex and HF subtype, our dataset offers a structured starting point for exploring sex-disease interactions in-silico. Furthermore, it can serve as a foundation for synthetic cohort generation [17, 18] with a more diverse range of anatomical and clinical characteristics. Secondly, the simulations performed in this work were benchmark simulations and the parameters were not calibrated to the individual anatomies. However, the use of a common set of parameters across all cases ensured that the differences observed in the results were purely anatomical. Third, the pipeline requires 5 hours per heart on average, with 2 hours spent performing manual corrections required at the segmentation stage. Even so, this is a significant reduction compared to previous methods. Finally, limitations remain at the segmentation stage. Because this step underpins all subsequent model construction, we discuss its implications in more detail in the following section.

Our simulations employed uniform electrophysiological and mechanical parameters across all models to isolate the effects of anatomical variability. The positive correlation between simulated activation times and QRS duration in control hearts ($r = 0.649, p < 0.005$) indicates that reference conductivity parameters provide an approximate anatomical surrogate for QRS, and that anatomical variation captures at least part of the inter-subject variability in conduction time. It does not, however, constitute a precise prediction of QRS, nor does it account for patient-specific tissue properties such as fibrosis or conduction system disease. The absence of correlation in heart failure patients ($r = -0.127, p = 0.64$) reflects this limitation and highlights the need for patient-specific parameter calibration.

Additionally, the UKBB reference ranges used for anatomical comparison [54] were derived from a predominantly Caucasian population. Ethnicity data were not systematically collected for the present cohort, which is consistent with common clinical practice in Europe [69], but may limit the generalisability of comparisons with these reference ranges for non-Caucasian patients. Racial differences in cardiac structure and function have been reported [70], underscoring the importance of ethnicity-matched reference data when assessing anatomical fidelity in diverse cohorts.

5.5.1 Critical Role of Segmentation in Model Accuracy

The segmentation stage is a critical bottleneck in the pipeline, requiring notable manual input to ensure anatomical fidelity. The most frequent issues involve small gaps or discontinuities in the myocardium, where the wall becomes excessively thin. The segmentation method proposed by [29] is not robust against some artefacts introduced by implanted devices, such as the bright streaks caused by leads or stents in HF patients. Manual steps and corrections in the pipeline could potentially be replaced with more robust algorithms. For example, recent work by [71] demonstrated a foundation model capable of deriving full myocardial masks without user input. Finally, given the limited soft-tissue contrast and resolution of standard CT scans, the myocardial thickness in chambers outside of the LV is set to a user-specified constant. This approximation may not reflect the true myocardial thickness and may underestimate simulation variability. Full voxel-level correspondence assessment of the 37-label post-processed geometry against clinical images is not feasible for this dataset: CT soft-tissue contrast does not resolve myocardial boundaries outside the LV, and structures including the RV wall, atria, and great vessel walls are assigned thickness via distance maps at a user-specified constant. For the 10-label segmentation stage, geometric accuracy is supported by the published validation of the deep learning model [29]. Population-level consistency of chamber volumes with UKBB reference ranges (Section 3.3) provides indirect geometric plausibility evidence at the cohort level.

5.5.2 Uncertainty Quantification

This study addresses variability across the population — how anatomical differences between patients influence simulation outcomes under fixed parameters. A distinct and equally important source of variability arises during model construction, introduced through segmentation choices and boundary placement. Previous work has begun to quantify this construction-stage uncertainty in the context of left atrial models [**?**, 1], demonstrating that operator variability in segmentation introduces quantifiable differences in chamber geometry and downstream simulation predictions. However, whole-heart models present substantially greater complexity, with four chambers, multiple vessel junctions, and valve planes that may amplify uncertainty propagation through the reconstruction workflow. Explicitly quantifying construction-stage uncertainty in whole-heart models has not been addressed here and represents an important limitation, particularly in the context of regulatory credibility frameworks such as those outlined by the FDA [68].

Despite these limitations, our current approach balances physiological realism with scalability, enabling whole-heart simulations across a sizeable cohort. Incorporating more refined fibre estimation methods remains an important direction for improving predictive accuracy in personalised cardiac models. By incorporating this level of anatomical detail into a reproducible, Python-based workflow, our pipeline bridges the gap between clinical image segmentation and simulation-ready models. It provides a viable route to building whole-heart virtual cohorts for in-silico trials and aligns with regulatory guidelines on model credibility and anatomical completeness [12].

6 Conclusion

We presented a cohort of 50 publicly available whole-heart simulations derived from patient-specific CT geometries, comprising healthy individuals and HF patients balanced by sex. The dataset offers a representative and well-characterised virtual cohort for in-silico cardiac research. The cohort is suitable for evaluating modeling assumptions, exploring anatomical variability, and help the development of digital-twin–style pipelines contingent on parameter calibration, data assimilation, and validation. Our results show that anatomical variability alone can drive consistent differences between groups, underscoring the need to account for both sex- and disease-specific features when designing and interpreting simulation studies. By combining anatomical fidelity with a reproducible workflow, this work provides a practical foundation for larger-scale investigations and the clinical translation of whole-heart simulations.

Author Contributions

Authors' initials: José Alonso Solís-Lemus (JASL); Rosie K. Barrows (RKB); Cristobal Rodero (CRG); Marina Strocchi (MS); Natalie Montarello (NM); Nishant Lahoti (NL); Cesare Corrado (CC); Abdul Qayyum (AQ); Shahrokh Rahmani (SR); Caroline Roney (CR); Gernot Plank (GP); Cristoph Augustin (CA); Hao Xu (HX); Alistair Young (AY); Pras Pathmanathan (PP); Ronak Rajani (RR); Steven A. Niederer (SN).

Conceptualization: JASL, RKB, MS, PP, SN. **Data curation:** NM, NL, RR. **Formal analysis:** JASL, RKB. **Funding acquisition:** CR, GP, CA, AY, SN. **Investigation:** JASL, RKB. **Methodology:** JASL, RKB, CRG, MS, CC, AQ, SR, CR, GP, CA, HX. **Project administration:** SN. **Resources:** NM, NL, RR. **Software:** JASL, RKB, CRG, MS. **Supervision:** AY, RR, PP, SN. **Validation:** JASL, RKB. **Visualization:** JASL, RKB. **Writing – original draft:** JASL, RKB, CRG, PP. **Writing – review & editing:** All authors.

References

1. Solís-Lemus JA, Baptiste T, Barrows R, Sillett C, Gharaviri A, Raffaele G, et al. Evaluation of an open-source pipeline to create patient-specific left atrial models: A reproducibility study. Computers in Biology and Medicine. 2023;162:107009. doi:10.1016/j.compbiomed.2023.107009.

2. Gray RA, Pathmanathan P. Patient-Specific Cardiovascular Computational Modeling: Diversity of Personalization and Challenges. Journal of Cardiovascular Translational Research. 2018 Apr;11(2):80-8. doi:10.1007/s12265-018-9792-2.

3. Vanderlaan RD, Nardo AD, Amon CH. A Multidisciplinary Approach to Patient-Specific Surgical Planning in Congenital Heart Disease. JACC: Advances. 2024;3(7_Part_1):101058. doi:10.1016/j.jacadv.2024.101058.

4. Niederer SA, Lumens J, Trayanova NA. Computational models in cardiology. Nature reviews Cardiology. 2019 Feb;16(2):100-11. doi:10.1038/s41569-018-0104-y.

5. Sung E, Kyranakis S, Daimee UA, Engels M, Prakosa A, Zhou S, et al. Evaluation of a deep learning-enabled automated computational heart modelling workflow for personalized assessment of ventricular arrhythmias. The Journal of Physiology. 2024 Sep;602(18):4625-44. doi:10.1113/JP284125.

6. Lee AWC, Costa CM, Strocchi M, Rinaldi CA, Niederer SA. Computational Modeling for Cardiac Resynchronization Therapy. Journal of Cardiovascular Translational Research. 2018;11(2):92-108. doi:10.1007/s12265-017-9779-4.

7. Dokuchaev A, Chumarnaya T, Bazhutina A, Khamzin S, Lebedeva V, Lyubimtseva T, et al. Combination of personalized computational modeling and machine learning for optimization of left ventricular pacing site in cardiac resynchronization therapy. Frontiers in Physiology. 2023 Jul;14. Publisher: Frontiers. doi:10.3389/fphys.2023.1162520.

8. Morrison TM, Pathmanathan P, Adwan M, Margerrison E. Advancing Regulatory Science With Computational Modeling for Medical Devices at the FDA's Office of Science and Engineering Laboratories. Frontiers in Medicine. 2018;5. doi:10.3389/fmed.2018.00241.

9. Jaffery OA, Melki L, Slabaugh G, Good WW, Roney CH. A Review of Personalised Cardiac Computational Modelling Using Electroanatomical Mapping Data. Arrhythmia Electrophysiology Review 2024;13:e08. 2024. doi:10.15420/aer.2023.25.

10. Shepard T, Scott G, Cole S, Nordmark A, Bouzom F. Physiologically Based Models in Regulatory Submissions: Output From the ABPI/MHRA Forum on Physiologically Based Modeling and Simulation. CPT: Pharmacometrics & Systems Pharmacology. 2015;4(4):221-5. doi:https://doi.org/10.1002/psp4.30.

11. Agency EM. EMA Regulatory Science to 2025. Strategic reflection;. Publisher: European Medicines Agency (EMA). Available from: `www.ema.europa.eu/en/documents/regulatory-procedural-guideline/ema-regulatory-science-2025-strategic-reflection_en.pdf`.

12. Health CfDaR. Credibility of Computational Models Program: Research on Computational Models and Simulation Associated with Medical Devices; 2023. Publisher: FDA. Available from: `www.fda.gov/medical-devices/medical-device-regulatory-science-research-programs-conducted-osel/credibility-computational-models-program-research-computational-models-an`

13. Strocchi M, Augustin CM, Gsell MAF, Karabelas E, Neic A, Gillette K, et al. A publicly available virtual cohort of four-chamber heart meshes for cardiac electro-mechanics simulations. PLOS ONE. 2020;15(6):e0235145. Publisher: Public Library of Science. doi:10.1371/journal.pone.0235145.

14. Rodero C, Strocchi M, Marciniak M, Longobardi S, Whitaker J, O'Neill MD, et al. Linking statistical shape models and simulated function in the healthy adult human heart. PLOS Computational Biology. 2021 04;17(4):1-28. doi:10.1371/journal.pcbi.1008851.

15. Roney C, Sim I, Yu J, Beach M, Mehta A, Solis-Lemus J, et al. Predicting atrial fibrillation recurrence by combining population data and virtual cohorts of patient-specific left atrial models. Circulation Arrhythmia and electrophysiology. 2022 Jan.

16. Passini E, Zhou X, Trovato C, Britton OJ, Bueno-Orovio A, Rodriguez B. The virtual assay software for human *in silico* drug trials to augment drug cardiac testing;52:101202. doi:10.1016/j.jocs.2020.101202.

17. Rodero C, Strocchi M, Marciniak M, Longobardi S, Whitaker J, O'Neill MD, et al.. Virtual cohort of 1000 synthetic heart meshes from adult human healthy population. Zenodo; 2021. Available from: `https://doi.org/10.5281/zenodo.4506930`. doi:10.5281/zenodo.4506930.

18. Rodero C, Strocchi M, Marciniak M, Longobardi S, Whitaker J, O'Neill MD, et al.. Virtual cohort of extreme and average four-chamber heart meshes from statistical shape model. Zenodo; 2021. Available from: `https://doi.org/10.5281/zenodo.4593739`. doi:10.5281/zenodo.4593739.

19. Feng L, Gao H, Luo X. Whole-heart modelling with valves in a fluid–structure interaction framework;420:116724. doi:10.1016/j.cma.2023.116724.

20. Okenov A, Nezlobinsky T, Zeppenfeld K, Vandersickel N, Panfilov AV. Computer based method for identification of fibrotic scars from electrograms and local activation times on the epi- and endocardial surfaces of the ventricles;19(4):1-21. Publisher: Public Library of Science. doi:10.1371/journal.pone.0300978.

21. Kaboudian A, Cherry EM, Fenton FH. GPU Load Balancing Using Sparse Cartesian Grids: Making Interactive WebGL Simulations of Complex Ionic Models Even Faster on 3D Heart Structures. In: 2023 Computing in Cardiology (CinC). vol. 50;. p. 1-4. ISSN: 2325-887X. doi:10.22489/CinC.2023.136.

22. Steuernagel CR, Lam CSP, Greenhalgh T. Countering sex and gender bias in cardiovascular research requires more than equal recruitment and sex disaggregated analyses. BMJ. 2023 Aug;382:e075031. Publisher: British Medical Journal Publishing Group Section: Analysis. doi:10.1136/bmj-2023-075031.

23. Pak DH, Liu M, Kim T, Liang L, Caballero A, Onofrey J, et al. Patient-Specific Heart Geometry Modeling for Solid Biomechanics Using Deep Learning. IEEE Transactions on Medical Imaging. 2024;43(1):203-15. Conference Name: IEEE Transactions on Medical Imaging. doi:10.1109/TMI.2023.3294128.

24. Razeghi O, Solís-Lemus JA, Lee AWC, Karim R, Corrado C, Roney CH, et al. CemrgApp: An interactive medical imaging application with image processing, computer vision, and machine learning toolkits for cardiovascular research. SoftwareX. 2020 Jul;12:100570. doi:10.1016/j.softx.2020.100570.

25. Sim I, Razeghi O, Karim R, Chubb H, Whitaker J, O'Neill L, et al. Reproducibility of atrial fibrosis assessment using CMR imaging and an open source platform. JACC: Cardiovascular Imaging. 2019 Oct;12(10):2076-7. doi:10.1016/j.jcmg.2019.03.027.

26. Sillett C, Razeghi O, Lee AWC, Solis Lemus JA, Roney C, Mannina C, et al. A three-dimensional left atrial motion estimation from retrospective gated computed tomography: application in heart failure patients with atrial fibrillation. Frontiers in Cardiovascular Medicine. 2024;11. Publisher: Frontiers. doi:10.3389/fcvm.2024.1359715.

27. Sillett C, Razeghi O, Lee A, Solis Lemus JA, Roney C, Ganesan P, et al. Novel Regional Analysis of Left Atrial Strain From Computed Tomography Separates Patients With Persistent versus Paroxysmal Atrial Fibrillation. Circulation. 2023;148:A13321-1. Publisher: American Heart Association. doi:10.1161/circ.148.suppl_1.13321.

28. Vigmond EJ, Hughes M, Plank G, Leon LJ. Computational tools for modeling electrical activity in cardiac tissue. Journal of Electrocardiology. 2003;36:69-74. doi:https://doi.org/10.1016/j.jelectrocard.2003.09.017.

29. Xu H, Niederer SA, Williams SE, Newby DE, Williams MC, Young AA. Whole Heart Anatomical Refinement from CCTA using Extrapolation and Parcellation. arXiv;. arXiv:2111.09650 [eess]. doi:10.48550/arXiv.2111.09650.

30. Crozier A, Augustin CM, Neic A, Prassl AJ, Holler M, Fastl TE, et al. Image-Based Personalization of Cardiac Anatomy for Coupled Electromechanical Modeling;44(1):58-70. Available from: https://doi.org/10.1007/s10439-015-1474-5. doi:10.1007/s10439-015-1474-5.

31. CGAL 4.11 - Manual: Package Overview;. Available from: https://doc.cgal.org/4.11/Manual/packages.html.

32. Neic A, Gsell MAF, Karabelas E, Prassl AJ, Plank G. Automating image-based mesh generation and manipulation tasks in cardiac modeling workflows using Meshtool. SoftwareX. 2020;11:100454. doi:https://doi.org/10.1016/j.softx.2020.100454.

33. Bayer J, Prassl AJ, Pashaei A, Gomez JF, Frontera A, Neic A, et al. Universal ventricular coordinates: A generic framework for describing position within the heart and transferring data. Medical Image Analysis. 2018;45:83-93. Available from: https://www.sciencedirect.com/science/article/pii/S1361841518300203. doi:https://doi.org/10.1016/j.media.2018.01.005.

34. Strocchi M, Gsell MAF, Augustin CM, Razeghi O, Roney CH, Prassl AJ, et al. Simulating ventricular systolic motion in a four-chamber heart model with spatially varying robin boundary conditions to model the effect of the pericardium. Journal of Biomechanics. 2020 Mar;101:109645. doi:10.1016/j.jbiomech.2020.109645.

35. Labarthe S, Bayer J, Coudière Y, Henry J, Cochet H, Jaïs P, et al. A bilayer model of human atria: mathematical background, construction, and assessment. EP Europace. 2014 11;16(suppl 4):iv21-9. doi:10.1093/europace/euu256.

36. Roney CH, Bendikas R, Pashakhanloo F, Corrado C, Vigmond EJ, McVeigh ER, et al. Constructing a Human Atrial Fibre Atlas. Annals of Biomedical Engineering. 2021 Jan;49(1):233-50. doi:10.1007/s10439-020-02525-w.

37. Bayer JD, Blake RC, Plank G, Trayanova NA. A Novel Rule-Based Algorithm for Assigning Myocardial Fiber Orientation to Computational Heart Models;40(10):2243-54. doi:10.1007/s10439-012-0593-5.

38. Durrer D, Van Dam RT, Freud GE, Janse MJ, Meijler FL, Arzbaecher RC. Total Excitation of the Isolated Human Heart. Circulation. 1970 Jun;41(6):899-912. Publisher: American Heart Association. doi:10.1161/01.CIR.41.6.899.

39. Gillette K, Gsell MAF, Bouyssier J, Prassl AJ, Neic A, Vigmond EJ, et al. Automated Framework for the Inclusion of a His-Purkinje System in Cardiac Digital Twins of Ventricular Electrophysiology. Annals of Biomedical Engineering. 2021 Dec;49(12):3143-53. doi:10.1007/s10439-021-02825-9.

40. Gillette K, Gsell MAF, Prassl AJ, Karabelas E, Reiter U, Reiter G, et al. A Framework for the generation of digital twins of cardiac electrophysiology from clinical 12-leads ECGs. Medical Image Analysis. 2021;71:102080. Available from: https://www.sciencedirect.com/science/article/pii/S1361841521001262. doi:https://doi.org/10.1016/j.media.2021.102080.

41. Neic A, Campos FO, Prassl AJ, Niederer SA, Bishop MJ, Vigmond EJ, et al. Efficient computation of electrograms and ECGs in human whole heart simulations using a reaction-eikonal model. Journal of Computational Physics. 2017;346:191-211. Available from: https://www.sciencedirect.com/science/article/pii/S0021999117304655. doi:https://doi.org/10.1016/j.jcp.2017.06.020.

42. Lee AWC, Nguyen UC, Razeghi O, Gould J, Sidhu BS, Sieniewicz B, et al. A rule-based method for predicting the electrical activation of the heart with cardiac resynchronization therapy from non-invasive clinical data. Medical Image Analysis. 2019;57:197-213. doi:https://doi.org/10.1016/j.media.2019.06.017.

43. Guccione JM, McCulloch AD, Waldman LK. Passive Material Properties of Intact Ventricular Myocardium Determined From a Cylindrical Model. Journal of Biomechanical Engineering. 1991 02;113(1):42-55. Available from: https://doi.org/10.1115/1.2894084. doi:10.1115/1.2894084.

44. Faulkner MG, Haddow JB. Nearly isochoric finite torsion of a compressible isotropic elastic circular cylinder. Acta Mechanica. 1972 Sep;13(3):245-53. Available from: https://doi.org/10.1007/BF01586796. doi:10.1007/BF01586796.

45. Strocchi M, Longobardi S, Augustin CM, Gsell MAF, Petras A, Rinaldi CA, et al. Cell to whole organ global sensitivity analysis on a four-chamber heart electromechanics model using Gaussian processes emulators. PLOS Computational Biology. 2023 Jun;19(6):e1011257. Publisher: Public Library of Science. doi:10.1371/journal.pcbi.1011257.

46. Sanders P, Morton JB, Davidson NC, Spence SJ, Vohra JK, Sparks PB, et al. Electrical Remodeling of the Atria in Congestive Heart Failure. Circulation. 2003;108(12):1461-8. doi:10.1161/01.CIR.0000090688.49283.67.

47. Buck S, Rienstra M, Maass AH, Nieuwland W, Van Veldhuisen DJ, Van Gelder IC. Cardiac resynchronization therapy in patients with heart failure and atrial fibrillation: importance of new-onset atrial fibrillation and total atrial conduction time. EP Europace. 2008 03;10(5):558-65. doi:10.1093/europace/eun064.

48. Cohen J. Statistical power analysis for the behavioral sciences. routledge; 2013.

49. Benjamini Y, Hochberg Y. Controlling the false discovery rate: a practical and powerful approach to multiple testing. Journal of the royal statistical society series b-methodological. 1995;57:289-300.

50. Gignac GE, Szodorai ET. Effect size guidelines for individual differences researchers. Personality and Individual Differences. 2016 Nov;102:74-8. doi:10.1016/j.paid.2016.06.069.

51. Rossi A, Cicoira M, Bonapace S, Golia G, Zanolla L, Franceschini L, et al. Left atrial volume provides independent and incremental information compared with exercise tolerance parameters in patients with heart failure and left ventricular systolic dysfunction. Heart. 2007;93(11):1420-5. doi:10.1136/hrt.2006.101261.

52. Kasa G, Teis A, Juncà G, Aimo A, Lupón J, Cediel G, et al. Clinical and prognostic implications of left ventricular dilatation in heart failure. European Heart Journal - Cardiovascular Imaging. 2024 01;25(6):849-56. doi:10.1093/ehjci/jeae025.

53. Ito K, Li S, Homma S, Thompson JLP, Buchsbaum R, Matsumoto K, et al. Left ventricular dimensions and cardiovascular outcomes in systolic heart failure: the WARCEF trial. ESC Heart Failure. 2021;8(6):4997-5009. doi:10.1002/ehf2.13560.

54. Petersen SE, Aung N, Sanghvi MM, Zemrak F, Fung K, Paiva JM, et al. Reference ranges for cardiac structure and function using cardiovascular magnetic resonance (CMR) in Caucasians from the UK Biobank population cohort;19(1):18. doi:10.1186/s12968-017-0327-9.

55. Roney CH, Solis Lemus JA, Lopez Barrera C, Zolotarev A, Ulgen O, Kerfoot E, et al. Constructing bilayer and volumetric atrial models at scale. Interface Focus. 2023 Dec;13(6):20230038. Publisher: Royal Society. doi:10.1098/rsfs.2023.0038.

56. Azzolin L, Eichenlaub M, Nagel C, Nairn D, Sánchez J, Unger L, et al. AugmentA: Patient-specific augmented atrial model generation tool. Computerized Medical Imaging and Graphics: The Official Journal of the Computerized Medical Imaging Society. 2023 Sep;108:102265. doi:10.1016/j.compmedimag.2023.102265.

57. Govil S, Crabb BT, Deng Y, Dal Toso L, Puyol-Antón E, Pushparajah K, et al. A deep learning approach for fully automated cardiac shape modeling in tetralogy of Fallot. Journal of Cardiovascular Magnetic Resonance: Official Journal of the Society for Cardiovascular Magnetic Resonance. 2023 Feb;25(1):15. doi:10.1186/s12968-023-00924-1.

58. Banerjee A, Camps J, Zacur E, Andrews CM, Rudy Y, Choudhury RP, et al. A completely automated pipeline for 3D reconstruction of human heart from 2D cine magnetic resonance slices. Philosophical Transactions Series A, Mathematical, Physical, and Engineering Sciences. 2021 Dec;379(2212):20200257. doi:10.1098/rsta.2020.0257.

59. Zhang Y, Liang X, Ma J, Jing Y, Gonzales MJ, Villongco C, et al. An Atlas-Based Geometry Pipeline for Cardiac Hermite Model Construction and Diffusion Tensor Reorientation. Medical image analysis. 2012 Aug;16(6):1130-41. doi:10.1016/j.media.2012.06.005.

60. Gillette K, Gsell MAF, Strocchi M, Grandits T, Neic A, Manninger M, et al. A personalized real-time virtual model of whole heart electrophysiology. Frontiers in Physiology. 2022 Sep;13. Publisher: Frontiers. doi:10.3389/fphys.2022.907190.

61. MICCAI. WHS++: Whole Heart Segmentation Dataset. Available from: `http://www.zmic.org.cn/care_2024/track5/`.

62. Ro SK, Sato K, Ijuin S, Sela D, Fior G, Heinsar S, et al. Assessment and diagnosis of right ventricular failure—retrospection and future directions. Frontiers in Cardiovascular Medicine. 2023;Volume 10 - 2023. doi:10.3389/fcvm.2023.1030864.

63. Zile MR, Baicu CF, Gaasch WH. Diastolic Heart Failure — Abnormalities in Active Relaxation and Passive Stiffness of the Left Ventricle. New England Journal of Medicine. 2004;350(19):1953-9. doi:10.1056/NEJMoa032566.

64. Aurigemma GP, Zile MR, Gaasch WH. Contractile Behavior of the Left Ventricle in Diastolic Heart Failure. Circulation. 2006;113(2):296-304. doi:10.1161/CIRCULATIONAHA.104.481465.

65. Carpenito M, Fanti D, Mega S, Benfari G, Bono MC, Rossi A, et al. The Central Role of Left Atrium in Heart Failure. Frontiers in Cardiovascular Medicine. 2021;Volume 8 - 2021.

66. Kim D, Seo JH, Choi KH, Lee SH, Choi JO, Jeon ES, et al. Prognostic Implications of Left Atrial Stiffness Index in Heart Failure Patients With Preserved Ejection Fraction. JACC: Cardiovascular Imaging. 2023;16(4):435-45.

67. Pathmanathan P, Aycock K, Badal A, Bighamian R, Bodner J, Craven BA, et al. Credibility assessment of in silico clinical trials for medical devices. PLOS Computational Biology. 2024 Aug;20(8):e1012289. Publisher: Public Library of Science. doi:10.1371/journal.pcbi.1012289.

68. Health CfDaR. Assessing the Credibility of Computational Modeling and Simulation in Medical Device Submissions; 2023. Publisher: FDA. Available from: `https://www.fda.gov/regulatory-information/search-fda-guidance-documents/assessing-credibility-computational-modeling-and-simulation-medical-devic`

69. Sheikh A, Netuveli G, Kai J, Panesar SS. Comparison of reporting of ethnicity in US and European randomised controlled trials. BMJ. 2004;329(7457):87-8. doi:10.1136/bmj.38061.593935.F7.

70. LaBounty TM, Bach DS, Bossone E, Kolias TJ. Effect of Race on Echocardiographic Measures of Cardiac Structure and Function. The American Journal of Cardiology. 2019;124(5):812-8. doi:https://doi.org/10.1016/j.amjcard.2019.05.049.

71. Qayyum A, Mazher M, Ugurlu D, Lemus JAS, Rodero C, Niederer SA. Foundation Model for Whole-Heart Segmentation: Leveraging Student-Teacher Learning in Multi-Modal Medical Imaging; 2025. Available from: `https://arxiv.org/abs/2503.19005`. arXiv:2503.19005.

Supporting Information

Supporting Information

S0 Figure. Graphical Abstract. Cohort creation and pipeline overview. Starting from clinical CT scans, the workflow performs multi-label segmentation and mesh generation to create simulation-ready models with assigned fibre orientations. A new dataset of 50 patients (26 controls, 24 with heart failure subtypes: narrow QRS and wide QRS) balanced by sex is processed. Resulting models are used in simulations of cardiac electrophysiology and mechanics to extract clinically relevant measurements.

S1 Figure. Correlation between simulated ventricular total activation time (TAT_V) and recorded ECG QRS duration, stratified by condition. Control hearts (blue, $n = 17$) showed strong positive correlation (Pearson $r = 0.649$, $p = 0.005$), validating TAT_V as a QRS surrogate when anatomical variability dominates. Heart failure patients (red, $n = 16$) showed no correlation ($r = -0.127$, $p = 0.64$), reflecting unmeasured pathological tissue properties.

S1 Table. Geometric characteristics stratified by sex and heart failure status. Mean ± SD reported for age (years), chamber volumes (mL), and total mesh elements (millions).

S2 Figure. Correlation matrix between geometric variables and simulation outputs. Spearman correlation coefficients (ρ) between geometric metrics (chamber volumes, mesh element count, age) and simulation outputs (activation times, volume changes). All shown correlations significant at $p < 0.05$.

S2 Table. Complete specification of CGAL meshing parameters used for volumetric mesh generation, including `facet_size`, `cell_size`, and `facet_distance` values.

S3 Table. List of all 37 anatomical labels assigned during segmentation and mesh generation, including label ID, anatomical structure name, and structure type.

S3 Figure. Distribution of mean element quality across the 50-heart cohort. Each point represents one mesh. The `tet_qmetric_volume` metric ranges from 0 (perfect tetrahedron) to 1 (degenerate); lower values indicate better quality. Cohort-wide mean was 0.153 ± 0.100. No inverted elements (quality > 0.99) were found in any mesh.

S1 Text. Detailed specification of pericardial constraints and boundary conditions applied in passive inflation simulations.

S1 Data. Per-model mesh quality statistics for all 50 hearts. Columns: mesh ID, total element count, min/max/mean/SD of `tet_qmetric_volume`, number of inverted elements, and percentage of near-degenerate elements.

S2 Data. Summary statistics for geometric characteristics and simulation outputs, stratified by sex and heart failure status.

S3 Data. Complete ANOVA results for all simulation outputs and geometric metrics.

S4 Data. All pairwise post-hoc comparison results (FDR-corrected) for simulation outputs and geometric metrics. Includes Cohen's d effect sizes and 95% confidence intervals.

S5 Data. Descriptive statistics (mean ± SD) for geometric variables stratified by group (Male/Female × Control/HF).

S6 Data. Spearman correlation coefficients between geometric variables and simulation outputs, with p-values and significance levels.